Guojun Xu

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Research Profile

I study learned distributed image compression at extremely low bitrates, with side information available only at the decoder. My work examines how generative reconstruction can balance perceptual quality and fidelity to the target image, using multimodal conditioning and explicit modeling of residual uncertainty. I am seeking an overseas joint Ph.D. research opportunity through the China Scholarship Council (CSC) application route.

Education

Wuhan University of Technology Sep. 2024 – Present

Direct-entry Ph.D. in Computer Science and Technology
Advisor: Junwei Zhou

Wuhan University of Technology Sep. 2020 – Jun. 2024

B.S. in Information and Computing Science

Publications

[1] **Guojun Xu**, Mingyang Zhang, Jianwen Xiang, Cheng Tan, Yanchao Yang, and Junwei Zhou. "Distributed Image Compression with Multimodal Side Information at Extremely Low Bitrates." *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. **Accepted.**

[2] **Guojun Xu**, Cheng Tan, Jianwen Xiang, and Junwei Zhou. "CODIC: Distributed Image Compression with Distortion–Perception Co-Optimization." *IEEE Transactions on Multimedia*, 2026. doi:10.1109/TMM.2026.3718254.

Submitted Manuscripts

[3] **Guojun Xu**, Jianwen Xiang, Yaning Xie, Yanchao Yang, and Junwei Zhou. "Perceptual Distributed Image Compression via Controlled Stochastic Reconstruction." *Data Compression Conference (DCC)*, 2027. **Under review.**

[4] **Guojun Xu**, et al. "Conditional Residual Decomposition for Distributed Image Compression." *International Conference on Learning Representations (ICLR)*, 2027. **Under review.**

[5] **Guojun Xu**, Jianwen Xiang, Mingyang Zhang, Yaning Xie, and Junwei Zhou. "Retrieval-Augmented Distributed Image Compression with Multi-Reference Side Information." *IEEE Transactions on Image Processing*. **Under review.**

[6] **Guojun Xu**, Mingyang Zhang, Jianwen Xiang, Cheng Tan, Yanchao Yang, and Junwei Zhou. "Distributed Image Compression with Target-Consistent Multimodal Side Information at Extremely Low Bitrates." *IEEE Transactions on Circuits and Systems for Video Technology*. **Submitted.**

Research Contributions

Perception–distortion trade-offs [2, 3]

Analyzed local perception–distortion behavior under a Gaussian approximation, relating reconstruction scale and source–reconstruction correlation to the distortion cost of stochastic refinement (SPDIC). Developed complementary distortion-oriented and diffusion-based reconstruction branches with adaptive regional fusion (CODIC). These works address the same trade-off through statistical analysis and architecture design, respectively.

Multimodal conditioning and side-information reliability [1, 6]

Developed a diffusion-based decoder that combines text derived from correlated images with visual side features. A multimodal alignment task supervises feature masks to exploit object-level category cues and fine details. Evaluated MDIC on KITTI Stereo and Cityscapes at target-image bitrates below 0.1 bits per pixel.

TCSVT extension: Investigating reliability-aware conditioning using bidirectional latent matching and gating with support, unknown, and conflict states. The method aims to limit unreliable conditioning when side features are mismatched or missing.

Conditional residuals and reconstruction uncertainty [4]

Applied conditional-expectation projection to separate the residual left by the target representation into a side-information-predictable component and an orthogonal remainder. Used the resulting variance partition to guide deterministic residual correction and stochastic modeling of the remaining uncertainty. This distinguishes content supported by the available conditions from variation supplied by the generative model.

Retrieval-based multi-reference compression [5]

Developed a complementary direction that retrieves references from a decoder-side gallery using the transmitted representation. Combined depth-aware geometric alignment and multi-reference cross-attention to handle differences in viewpoint and reference relevance, extending beyond a fixed paired-reference setting.

Prospective Research Directions

Reliable conditioning for generative compression with imperfect or retrieved side information; uncertainty-aware reconstruction that separates recoverable content from generated variation; evaluation of perceptual quality alongside target-image fidelity.

Awards and Scholarships

First-class Academic Scholarship for Ph.D. Students 2026

Wuhan University of Technology

Second-class Academic Scholarship for Ph.D. Students 2025

Wuhan University of Technology

First-class Academic Scholarship for Ph.D. Students 2024

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